Juan David Munoz Bolanos

Junior System Engineer

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Summary

Junior Engineer with research experience handling optomechatronic and medical devices from concept to prototype. Proven track record in translating physical concepts into functional prototypes using V-model systems engineering. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation for clinical and industrial applications.

Research Experience

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*
PhD Researcher in adaptive optics

End to End System Design: Led the hardware to software implementation of multiphoton microscope, integrating laser sources, galvo scanners, and light modulators. Achieved enhancing optical image quality.

Systems Engineering: instrument development following V-model principles, managing requirements and tracking from concept to component testing.

Quality & Testing: Executed Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies frameworks.

CAD & Prototyping: Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021 — **Leibniz Institute of Photonic Technology** — *Jena, Germany*
R&D Research Assistant

Spectrometer Testing: Testing and modifying a dispersive 1064 nm fiber probe Raman spectrometer for non-invasive human tissue characterization, handling optomechanical layout and detector calibration.

Data Analysis & ML: Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

Technical Documentation: Authored peer reviewed publications detailing optical design parameters and calibration procedures.

Technical Competencies

Hardware: Adaptive optics · Multiphoton/confocal microscopy · Raman spectroscopy · Laser manipulation · Camera integration · 3D CAD (SolidWorks, FreeCAD)

Software: Python (scikit-learn, NumPy) · LabVIEW/FPGA · C++ · OpenCV · Git · MATLAB

** Dark tags: expertise; light tags: knowledge.*

Education

Jan 2022 to Jun 2026 — **Medical University of Innsbruck** — *Innsbruck, Austria*
PhD in Adaptive Optics

Oct 2019 to Sep 2021 — **Friedrich Schiller Univ. Jena** — *Jena, Germany*
Master of Science in Photonics

Aug 2012 to Aug 2018	**University of Cauca** *Bachelor of Physics Engineering*	*Popayán, Colombia*

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Systems & Quality Engineering: V-Model, 8D Problem Solving, Design of Experiments.

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Biking/Hiking · Bachata Dancing · Acoustic Guitar · Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
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PD Dr. Christoph Krafft
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